

Catalytic Impact of Early Investments in Advanced Clean Energy Technologies

Kick-off

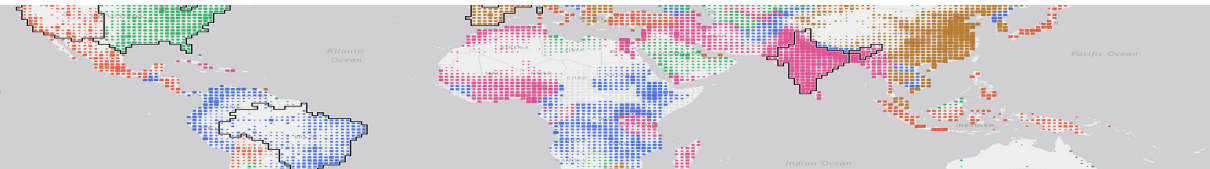
PyPSA Labs, Google

8 September 2026



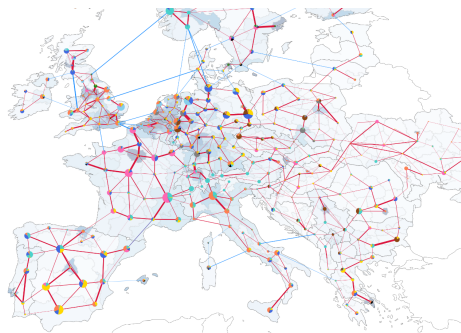
Project overview

- Lay the foundation for understanding the **catalytic impact of early investments** in advanced clean electricity technologies
- Quantitative illustration of the **direct and indirect** system-wide emissions reductions and cost-decline trajectories
- Inform **consequential and out-of-scope emissions accounting methodologies** and the broader conversation on how early-mover investments should be valued
- Deliver an **open-source, reproducible scientific workflow**



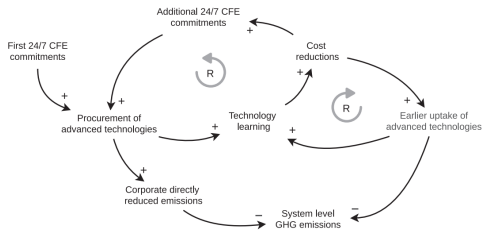
Modelling setup 1/2

- **PyPSA** for capacity expansion and dispatch optimisation
- **PyPSA-Earth** as the global data pipeline: brownfield fleet, demand, grid topology
- **atlite** for renewable resource potentials and capacity factors
- Representative regions *aka grid archetypes* featuring jurisdictions with varying grid topologies, renewable resource potentials, technology portfolios, etc.



Modelling setup 2/2

- Use 2025 as a **validation year**
- Run a **capacity-expansion model** for each grid archetype across investment years from 2025 to 2050
- Clear markets for advanced technologies between investment years; learning curves pass updated costs forward **myopically**
- Compare a **benchmark scenario** without catalytic action with a scenario including action by early investors
- Like-for-like comparison in system metrics (e.g. avoided emissions) to quantify impact of catalytic action



Scenarios

S1 — Reference pathway

Each archetype follows current cost trajectories and announced retirements, without catalytic investment. **Establishes the counterfactual baseline** for all impact estimates.

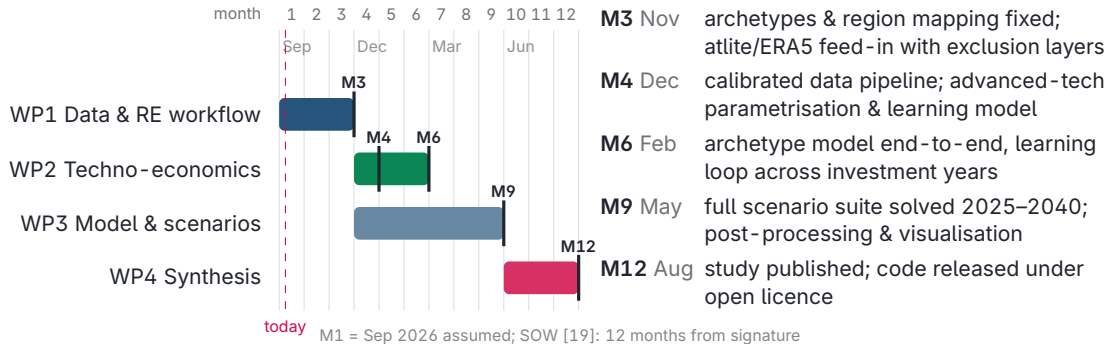
S2 — Catalytic early investment

Corporate and institutional buyers directly procure advanced clean technologies, with investment scale anchored to a share of regional demand and the portfolio optimised for local system needs. **Early deployment accelerates learning and cost competitiveness** beyond buyers' regions.

Additional scenarios represent alternative futures to baseline.

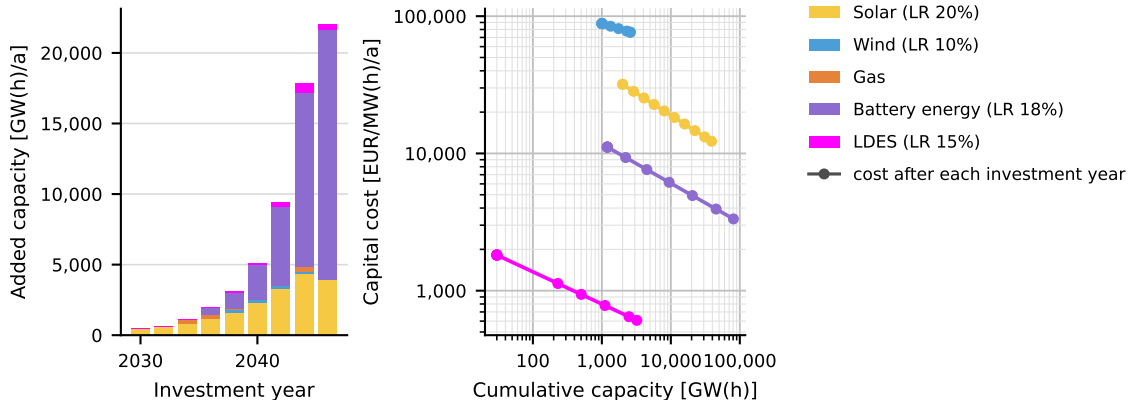
- **Wind/solar acceptance limits:** tighter land-use, protected-area and social-acceptance constraints increase the value of clean firm power.
- **CDR breakthrough:** engineered removal reaches about 150 EUR/tCO₂ by 2040, making gas plus CDR a competitive backup option.
- **Advanced baseload breakthrough:** rapid FOAK-to-commercial cost decline for SMRs or another firm clean technology.
- **Advanced geothermal breakthrough:** commercial success in geologically suitable regions creates locally significant catalytic impact.
- **SOFC breakthrough:** enable cheap, efficient backup running on hydrogen or natural gas with high CO₂ capture; reduces the premium for LDES and flexible clean generation.

SOW work packages and milestones



Prototype: Endogenous learning in the world model

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Open questions

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- Technology palette
- Initial investment mechanism
- Modelled regions

Technology palette

Mature: exogenous costs

- Solar PV, onshore/offshore wind
- Hydro: reservoir, run-of-river, pumped hydro
- Biomass
- Nuclear (large LWR)
- Gas CCGT and OCGT
- Li-ion battery

Advanced: endogenous learning

- LDES: CO₂ battery, iron-air battery, H₂ + SOFC, vanadium flow
- Nuclear SMR
- Next-gen geothermal: EGS, closed-loop
- Allam-cycle gas + CCS
- Solid oxide fuel cell
- CDR: direct air capture, cost cut-off

The Initial Investment Mechanism



- Generalise the initial action beyond 24/7 CFE to represent **multiple forms of catalytic investment**
- Let the model select the **locally appropriate technology portfolio**, rather than prescribing which technologies are procured
- Test two mechanisms with endogenous portfolio selection: a **demand-linked procurement target** and a **green-premium budget**

Modelled regions and grid archetypes



Renewables-constrained, high-density

☐ modelled: North-West Europe



Geothermal-favourable

☐ modelled: United States (West)



High renewables potential

☐ modelled: India



Hydro-rich

☐ modelled: Brazil



Fossil-heavy brownfield with large economies

☐ modelled: United States (East)

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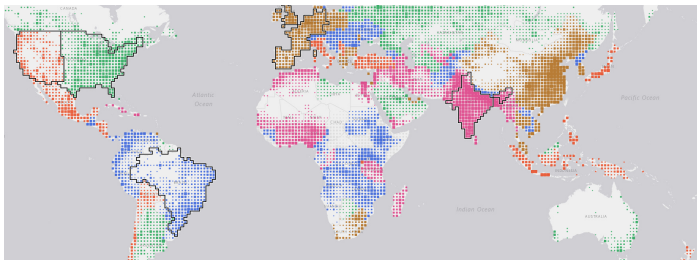
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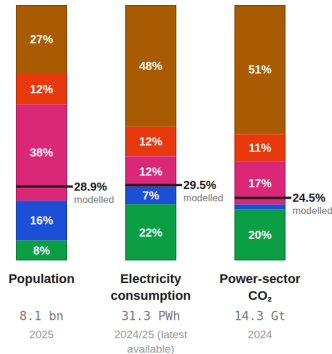
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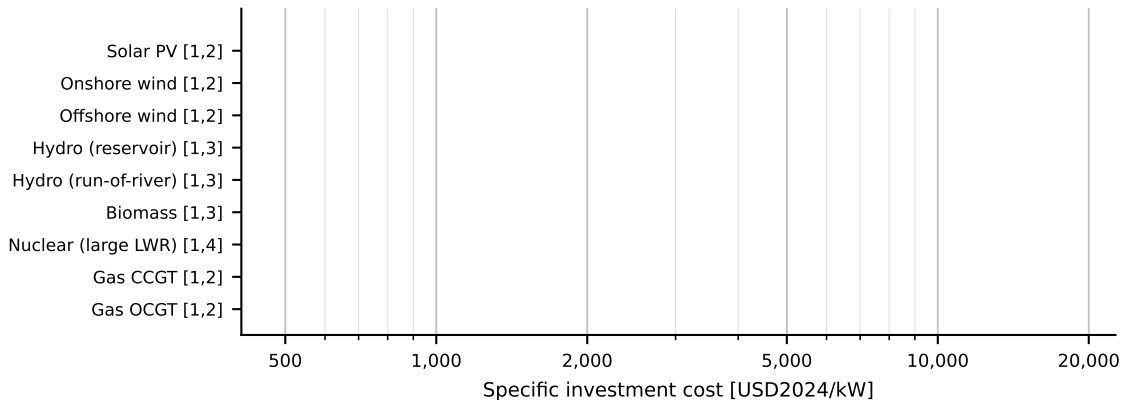
Fossil-heavy brownfield with large economies

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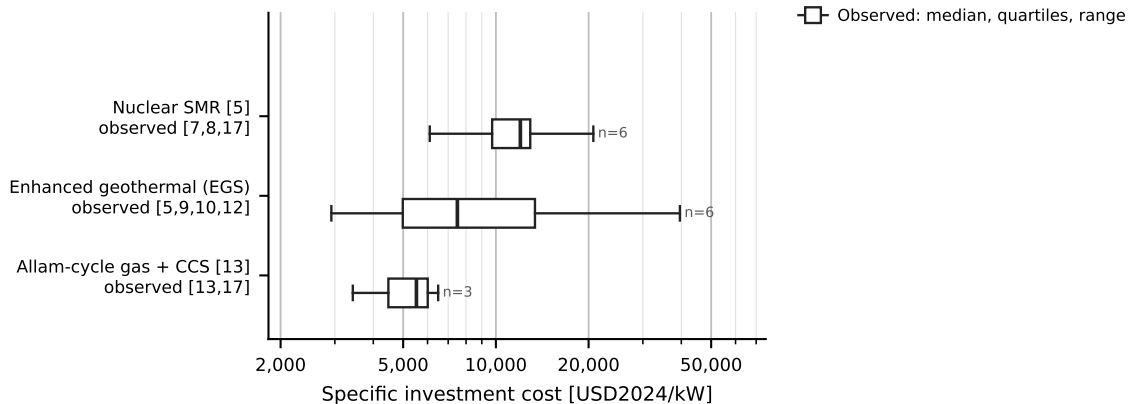


Annex

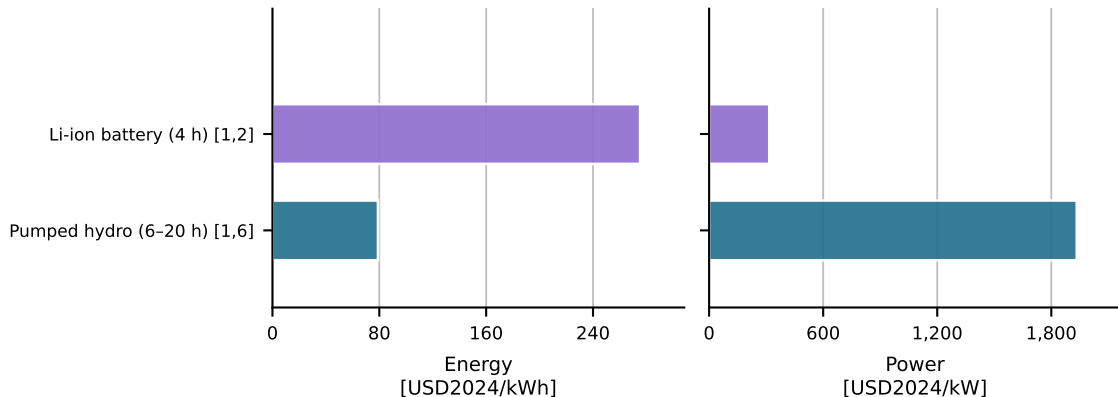
2025 capex baseline: mature generation



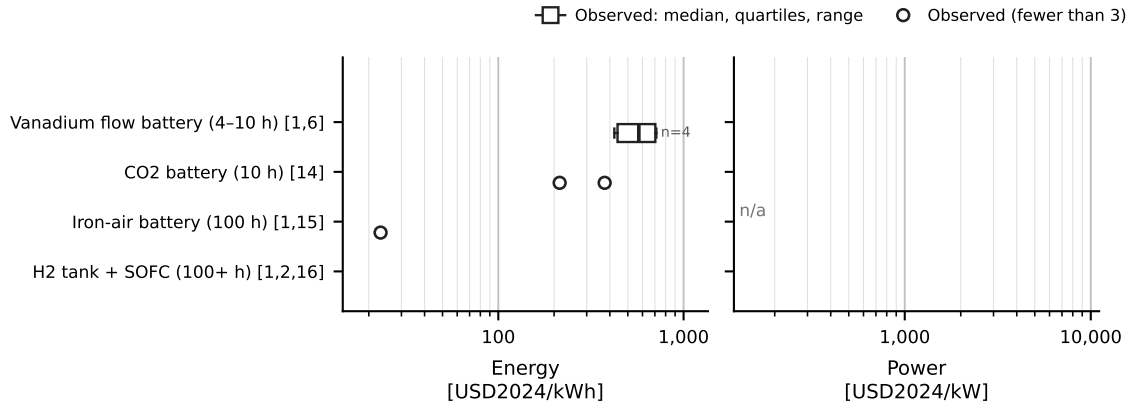
2025 capex baseline: advanced generation



2025 capex baseline: mature storage



2025 capex baseline: advanced storage



References

- [1] PyPSA, technology-data v0.15.0 (2025), cost assumptions for 2025 github.com
- [2] Danish Energy Agency, Technology Data catalogues: electricity generation; energy storage (2024/25) ens.dk
- [3] Schröder et al. (DIW Berlin), Costs of Electricity Generation until 2050, Data Documentation 68 (2013) diw.de
- [4] Lazard, Levelized Cost of Energy+ (LCOE+), June 2025 and v16.0 (2023) lazard.com
- [5] NREL, Annual Technology Baseline 2024, electricity (v3 data) atb.nrel.gov
- [6] PNNL / DOE, 2022 Grid Energy Storage Technology Cost and Performance Assessment (Viswanathan et al.) pnnl.gov
- [7] EIA / Sargent & Lundy, Capital Cost and Performance Characteristics for Utility-Scale Generating Technologies (AEO2025) eia.gov
- [8] US DOE, Pathways to Commercial Liftoff: Advanced Nuclear (Sep 2024 update) liftoff.energy.gov
- [9] US DOE, Pathways to Commercial Liftoff: Next-Generation Geothermal Power (Mar 2024) liftoff.energy.gov
- [10] NREL, Enhanced Geothermal Shot Analysis for the Geothermal Technologies Office (2023) nrel.gov
- [11] BloombergNEF, Energy Storage System Cost Survey 2024 (as reported by Energy-Storage.news) about.bnef.com
- [12] Fervo Energy, Form S-1 registration statement (2026), Cape Station cost disclosures sec.gov
- [13] NET Power, Q4 2024 results and business update (10 Mar 2025), Project Permian cost estimate ir.netpower.com
- [14] Energy Dome, CO2 Battery cost and performance statements (2022–2024) energydome.com
- [15] Form Energy, iron-air modelling recommendations (2023) and cost target statements formenergy.com
- [16] Bloom Energy, product cost disclosures 2017–2023 (via Thunder Said Energy); Energy Server datasheet (2026) bloomenergy.com
- [17] Developer, regulator and press disclosures of individual projects (Vogle, Hinkley Point C, Flamanville 3, Darlington, Kemmerer, Snowy 2.0, Fengning, Dalian, Alliant, ...); per-project sources in the repository github.com
- [18] T. Brown, priam-myopic: myopic world electricity model with endogenous learning (prototype, 2026) github.com
- [19] PyPSA Labs and Google, Statement of Work: Catalytic Impact of Early Investments in Advanced Clean Energy Tech (2026)

Get in touch

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